

HW1

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记投影中心 (小孔) O , 建立圆心为 C , 圆上任意两点 P_1, P_2 , 及其投影点 C', P'_1, P'_2
根据透视原理, $\triangle OCP_1 \sim \triangle OC'P'_1, \triangle OCP_2 \sim \triangle OC'P'_2$

$$\frac{OC}{OC'} = \frac{OP_1}{OP'_1} = \frac{OP_2}{OP'_2} \quad (1)$$

(2)

由于在圆中 $OP_1 = OP_2, OP'_1 = OP'_2$, 故而圆投影结果仍为一个圆, disk 的投影形状是圆形。

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Case 1: Horizontal Plane $y = 0$

Three different line directions in this plane are:

$$\mathbf{d}_1 = [1, 0, 0]$$

$$\mathbf{d}_2 = [0, 0, 1]$$

$$\mathbf{d}_3 = [1, 0, 1]$$

For a direction vector $\mathbf{d} = [x, y, z]$, the vanishing point (x_{vp}, y_{vp}) is given by:

$$x_{vp} = \frac{f \times x}{z}$$

$$y_{vp} = \frac{f \times y}{z}$$

For $\mathbf{d}_1$:

$$x_{vp1} = \frac{f \times 1}{0} = \infty$$

$$y_{vp1} = \frac{f \times 0}{0} = \text{undefined}$$

vp_1 is at infinity

For $\mathbf{d}_2$:

$$x_{vp2} = \frac{f \times 0}{1} = 0$$

$$y_{vp2} = \frac{f \times 0}{1} = 0$$

$$vp_2 = (0, 0)$$

For $\mathbf{d}_3$:

$$\begin{aligned}x_{vp3} &= \frac{f \times 1}{1} = f \\y_{vp3} &= \frac{f \times 0}{1} = 0 \\vp_3 &= (f, 0)\end{aligned}$$

Case 2: Vertical Plane $x = 0$

Three different line directions in this plane are:

$$\begin{aligned}\mathbf{d}_1 &= [0, 1, 0] \\ \mathbf{d}_2 &= [0, 0, 1] \\ \mathbf{d}_3 &= [0, 1, 1]\end{aligned}$$

For $\mathbf{d}_1$:

$$\begin{aligned}x_{vp1} &= \frac{f \times 0}{0} = \text{undefined} \\ y_{vp1} &= \frac{f \times 1}{0} = \infty \\ vp_1 &\text{is at infinity}\end{aligned}$$

For $\mathbf{d}_2$:

$$\begin{aligned}x_{vp2} &= \frac{f \times 0}{1} = 0 \\ y_{vp2} &= \frac{f \times 0}{1} = 0 \\ vp_2 &= (0, 0)\end{aligned}$$

For $\mathbf{d}_3$:

$$\begin{aligned}x_{vp3} &= \frac{f \times 0}{1} = 0 \\ y_{vp3} &= \frac{f \times 1}{1} = f \\ vp_3 &= (0, f)\end{aligned}$$

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Given a line with direction vector $\mathbf{d} = [x, y, z]$ in 3D space, the vanishing point of the line when projected onto the image plane can be found using the equations:

$$\begin{aligned}x_{vp} &= \frac{f \times x}{z} \\ y_{vp} &= \frac{f \times y}{z}\end{aligned}$$

where f is the effective focal length of the camera.

For a plane defined by the equation $Ax + By + Cz + D = 0$, z can be expressed in terms of x and y as:

$$z = -\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}$$

Given a direction vector $\mathbf{d}$ lying on this plane, its vanishing point can be determined by substituting the above expression for z into the equations for x_{vp} and y_{vp} :

$$x_{vp} = \frac{f \times x}{-\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}}$$

$$y_{vp} = \frac{f \times y}{-\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}}$$

These equations will give the vanishing points for all lines that have direction vectors $\mathbf{d}$ lying on the plane $Ax + By + Cz + D = 0$.